

GBENGA OMOTARA

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PhD student in Computer Vision and Robotics developing deep learning algorithms for perception tasks and multi-view scene understanding. Experienced in building end-to-end vision pipelines from data acquisition to deployment, with an emphasis on robust perception in challenging scenes.

EDUCATION

University of Missouri–Columbia

Columbia, MO

Ph.D., Electrical & Computer Engineering — Expected July 2026, ViGIR Lab

M.S., Computer Engineering (2023);

B.S., Electrical & Computer Engineering (2017).

RESEARCH & PROJECTS

EGSA-PT: Edge-Guided Spatial Attention with Progressive Training *(submitted)*

Developed a multi-task transformer for monocular depth estimation and segmentation of transparent objects. Introduced edge-guided spatial attention and progressive supervision, improving transparent-region geometry by 13% over baselines.

Vision-Language Models for Fine-grained Classification *CVPR 2025, WACV 2025*

Applied multiple vision–language models (VLMs) for VQA-based fine-grained image classification, treating VLM outputs as independent weak labelers within a Snorkel-based weak supervision framework.

Multi-View Cattle Reconstruction System *Sensors 2024*

Designed and deployed a synchronized 10-camera Azure Kinect rig for multi-view 3D capture. Implemented extrinsic calibration, custom sensor synchronization, TSDF fusion, and PCA-based surface measurements for volumetric growth estimation. Achieved <6mm alignment error across sessions.

WORK EXPERIENCE

University of Missouri — Vision Guided Intelligent Robotics (ViGIR) Lab *Columbia, MO*

Graduate Research Assistant

Jan 2018 – Present

- Conduct research on 3D reconstruction, monocular geometry estimation, and real-world perception.
- Developing multi-task geometric and semantic models toward real-time depth and segmentation for robotic perception.

EquipmentShare

Columbia, MO

Robotics / Machine Vision Intern

May 2022 – Jul 2022

- Built a real-time multi-sensor perception pipeline for heavy-equipment safety using stereo vision and RGB-D sensors.
- Reduced detection latency by 35% through an optimized parallel algorithm and asynchronous data streaming.
- Created a Unity-based simulation environment for sensor testing and pipeline validation under diverse lighting.

1. Omotara et al. Egsa-pt: Edge-guided spatial attention with progressive training for monocular depth estimation and segmentation of transparent objects. *Submitted to IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2026. Under review
2. Omotara et al. High-throughput and accurate 3d scanning of cattle using time-of-flight sensors and deep learning. *Sensors*, 24(16), 2024
3. Tousi et al. Combining vision-language models and weak supervision for nuanced vision classification tasks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pages 2142–2151, June 2025
4. Tousi et al. A zero-shot learning approach for ephemeral gully detection from remote sensing using vision language models. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) Workshops*, February 2025
5. Obukhov et al. The fourth monocular depth estimation challenge. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pages 6181–6194, June 2025
6. Farag et al. Efficientnet-sam: A novel efficientnet with spatial attention mechanism for covid-19 detection in pulmonary ct scans. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pages 5200–5206, June 2024
7. Omotara et al. A pipeline consisting of pattern recognition and finite automata for recognizing vcv productions in the study of vocal hyperfunction. In *2021 IEEE Symposium Series on Computational Intelligence (SSCI)*, pages 1–7, 2021
8. Chen et al. Partial membership latent dirichlet allocation for soft image segmentation. *IEEE Transactions on Image Processing*, 26(12):5590–5602, 2017

TECHNICAL SKILLS

Computer Languages :	Python, C/C++, MATLAB, Octave, Bash, L ^A T _E X
Software & Tools :	Kubernetes, Docker, Ollama, CloudCompare, MeshLab, PyTorch, Open3D, OpenCV, NVIDIA Jetson, Azure Kinect, Kinect v2, Intel RealSense, ZED 2, Visual Studio Code